Predicting Online Purchases

Project brief · DSA211 Topic 2: Classification System Evaluation

1. The problem

An online retailer wants to route live sales support (chat, assisted checkout) to the visitors most likely to buy *during* their session. We build and evaluate a classifier that predicts whether a browsing session ends in a purchase. A missed buyer (false negative) is a lost sale; a flagged non-buyer (false positive) is a wasted support slot. We treat a lost sale as five times as costly ($C_{FN}/C_{FP} = 5$) and will stress-test that assumption.

2. The dataset

Online Shoppers Purchasing Intention (UCI repository, dataset 468): 12,330 sessions from one Turkish online retailer over one year. The target **Revenue** records whether the session ended in a purchase. There are 17 predictors (9 numeric, 8 categorical) and no missing values.

Group	Variables	Example
Browsing behaviour (numeric)	pages visited and time spent, by page type (administrative, informational, product)	product pages viewed; seconds on them
Engagement (numeric)	average bounce rate and exit rate of the pages visited	share of visitors who leave from that page
Context (categorical)	month, closeness to a holiday, weekend, visitor type	new vs returning visitor
Technical (categorical)	operating system, browser, region, traffic source (anonymised codes)	traffic source 1–20

3. What the data tells us before modelling

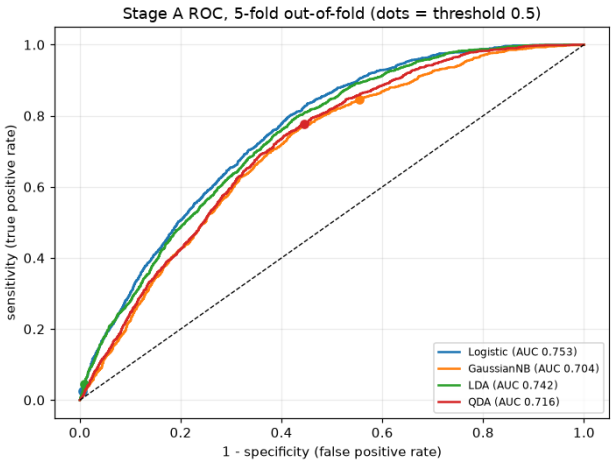
- **Imbalance.** Only 15.5% of sessions buy. Predicting "no purchase" for everyone would be 84.5% accurate and catch no buyers, so we evaluate with AUC, sensitivity and specificity, not accuracy.
- **Skew.** All 8 numeric behaviour features are strongly right-skewed in both classes (skewness up to 8.2), and informational-page features are 79–81% zeros. This breaks the bell-curve assumption behind Naive Bayes and LDA, so transformations are a core part of the plan.
- **Leakage.** One feature, *PageValues*, is computed by Google Analytics from sessions that completed a purchase. With it, cross-validated AUC is 0.894; without it, 0.753. That leakage premium of 0.141 would not survive deployment (82% of buyers have it non-zero, against 11% of non-buyers). We exclude it.

4. Baseline — four methods, no transformations, default settings

Evaluated by 25-fold cross-validation (5-fold × 5 repeats) on the training data only (9,247 sessions). The held-out test set (3,083 sessions) has not been touched and will be used once, at the end, for the single chosen model.

Model	CV AUC (±SD)	Sensitivity @ 0.5	Specificity @ 0.5	Brier score
Logistic regression	0.752 (±0.016)	0.025	0.994	0.118
LDA	0.741 (±0.015)	0.044	0.991	0.119
QDA *	0.715 (±0.014)	0.778	0.555	0.341
Naive Bayes	0.704 (±0.015)	0.846	0.446	0.452

* Default QDA could not be fitted: after one-hot encoding each class's covariance matrix is singular. A small regularisation (reg_param = 0.01) was needed.



Reading the baseline

Logistic regression ranks buyers best (AUC 0.752). Naive Bayes ranks worst (0.704), as its independence and normality assumptions predict.

Sensitivity is the problem. At the default 0.5 threshold, logistic regression catches only 2.5% of buyers. Naive Bayes appears to catch 85%, but only because its probabilities are badly miscalibrated (Brier 0.452 vs 0.118), not because it ranks better.

The median buyer receives a predicted probability of only 0.235; just 2.5% of buyers exceed 0.5, but 71.6% exceed the cost-optimal 0.167.

Figure: ROC curves from one 5-fold out-of-fold pass (dots mark the 0.5 threshold). The table reports the 25-fold mean, which is why the AUCs differ in the third decimal.

5. Why sensitivity is low — two causes, two kinds of fix

Cause	Fix we will test	Expected effect
A. A 0.5 threshold is mismatched to a 15.5% purchase rate	move the threshold; class weights; resampling (inside each fold)	**slides along** the ROC curve: sensitivity bought with specificity, AUC unchanged
B. The features carry limited signal	transformations, polynomial and interaction terms, domain features, tuning	**lifts** the ROC curve: more sensitivity at the same specificity, AUC rises

6. Plan

- **Assumption tests first.** Before transforming anything we test normality, equal covariance, independence, multicollinearity and linearity of the log-odds, so every later step targets a measured failure.
- **Stage B, transformations:** log, Yeo–Johnson, quantile, and zero-indicator transforms.
- **Stage C, nonlinearity:** polynomial and interaction terms, and QDA vs LDA. These ask the same question: does a curved boundary help?
- **Stage D, tuning:** regularisation strength (λ), subset size (at least five variables) and each method's smoothing.
- **Stage E, sensitivity:** the cause/fix/effect test above.
- **Ceiling benchmark:** gradient boosting, to show whether the limit is the data or the method.
- **Further checks:** KNN, splines and step functions, precision–recall, a bootstrap of the threshold, and a learning curve (would more data help?).
- **Finally:** choose one model on performance and on cost, justify a threshold other than 0.5 from training cross-validation, compare the CV estimate with the held-out test (used once), and refit on all data.

Every variant tried, including failures, is kept in a results ledger.

7. What we are trying to improve

Primary: cross-validated AUC. It measures ranking quality independently of any threshold; the baseline to beat is 0.752. **Operational:** sensitivity and specificity at a cost-justified threshold, where theory gives t* = 1/(1+5) = 0.167. Sensitivity is always reported together with specificity, since it alone can be pushed to 100% by flagging everyone. A real improvement is a better trade-off, not just a higher sensitivity.